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ICS 4111: Embedded Systems & IoT

Practical Exercise 3: Embedded System Design

Experiment 1

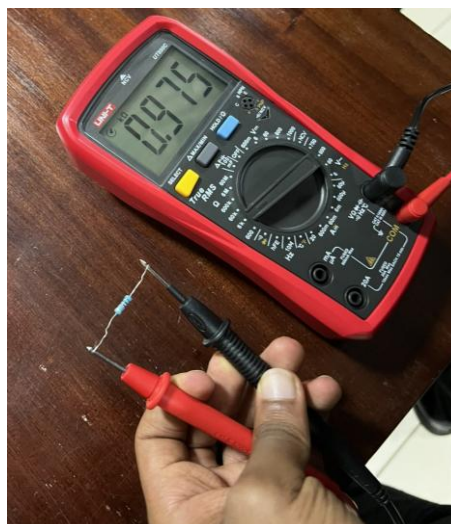
1. Pick one resistor, take a picture of it, and determine its resistance value based on the colour code. Write down the resistance value.



[We used this Resistor Colour Code Chart](#)

1 st Digit	2 nd Digit	Multiplier	Tolerance	Resistance
Brown (1)	Black (0)	Red ($\times 10^2$)	Gold ($\pm 5\%$)	= 1000 Ω $\pm$ 50 Ω

2. Using a multimeter, measure the resistance value emitted by the resistor. Take a picture of this setup and write down the measured resistance. You can check [this video](#) out or any other online resources on how to use a multimeter to measure voltage and resistance.



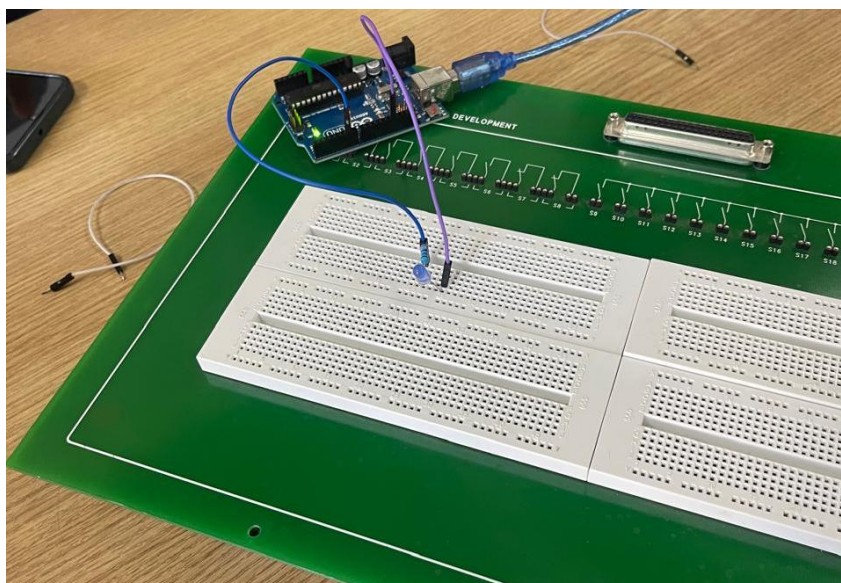
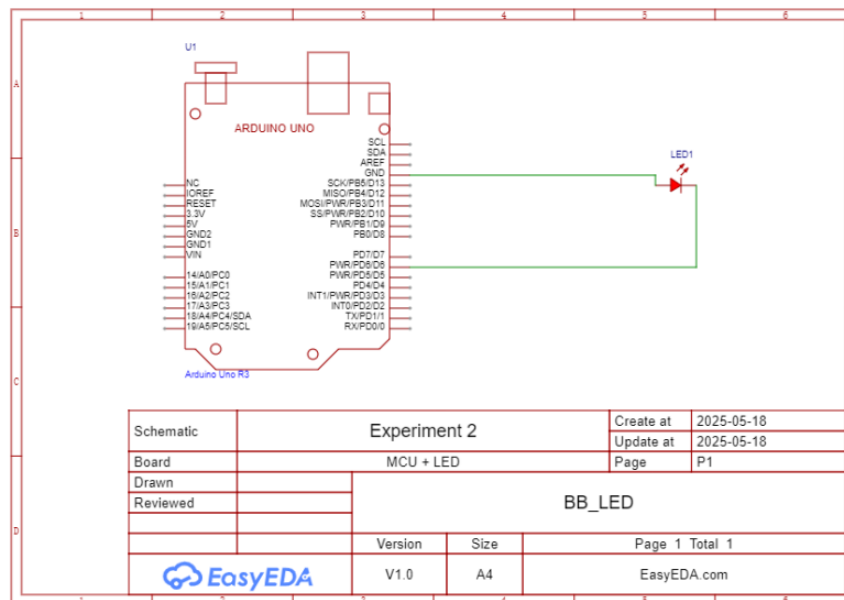
Measured Resistance, $R_1 = 0.975 \text{ k}\Omega = 975 \Omega$

Experiment Reflection: Where does the difference in the resistance values arise from?

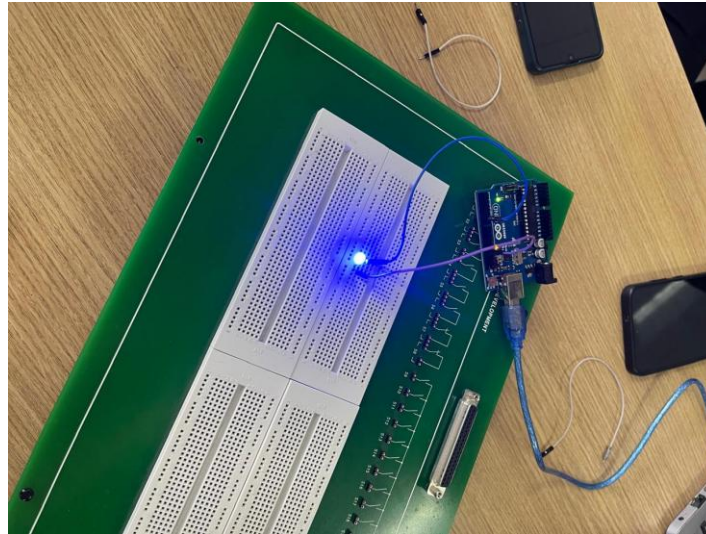
The difference arises from the manufacturing tolerance. This 1000Ω resistor is designed to operate within $\pm 5\%$ of its stated value, hence anything between 950Ω and 1050Ω is acceptable. The measured value 975Ω falls in this range. Other factors to consider include the accuracy of the multimeter, the small resistance of the multimeter's probes, and the temperature of the resistor.

Experiment 2

1. Using the breadboard, microcontroller, and LED, recreate the following schematic. Take a picture of the physical implementation.



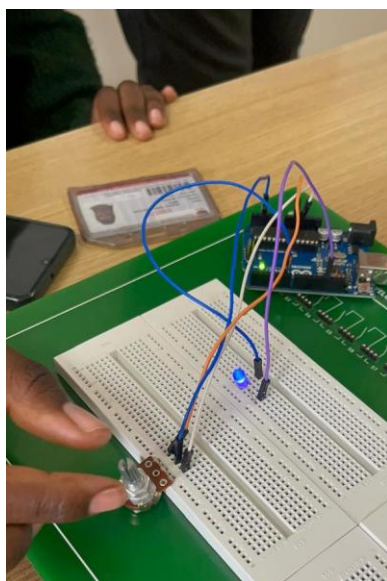
2. In Arduino IDE, deploy the [Blink](#) program which is usually inbuilt in Arduino and upload to your board. Congratulations you just created your first sketch! :)



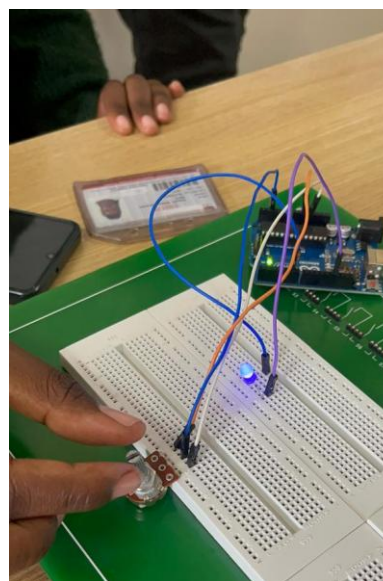
3. Thereafter, use the multimeter to determine the voltage passing through the circuit. Write down the value registered.

Measured Voltage, $V_1 = 4.8 \text{ V}$

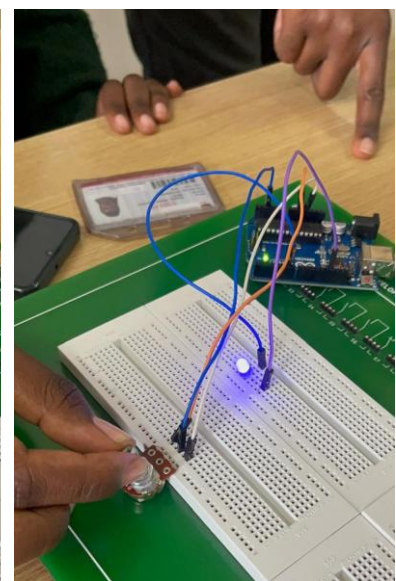
4. Reconfigure the circuit you created in question 1 by adding a potentiometer to simulate [Pulse Width Modulation \(PWM\)](#) on the LED. Take 3 images of your circuit with the LED at 25% brightness, 50% brightness and 100% brightness. You can refer to these or any other suitable resources to guide on the connection: [Using PWM to Control the Light Intensity of a LED - Instructables](#); [Arduino LED Dimmer \(Potentiometer + PWM\)](#).



25% Brightness



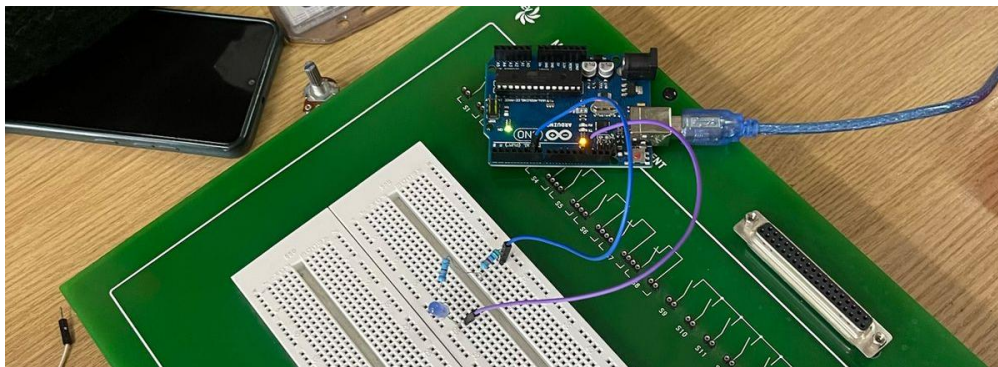
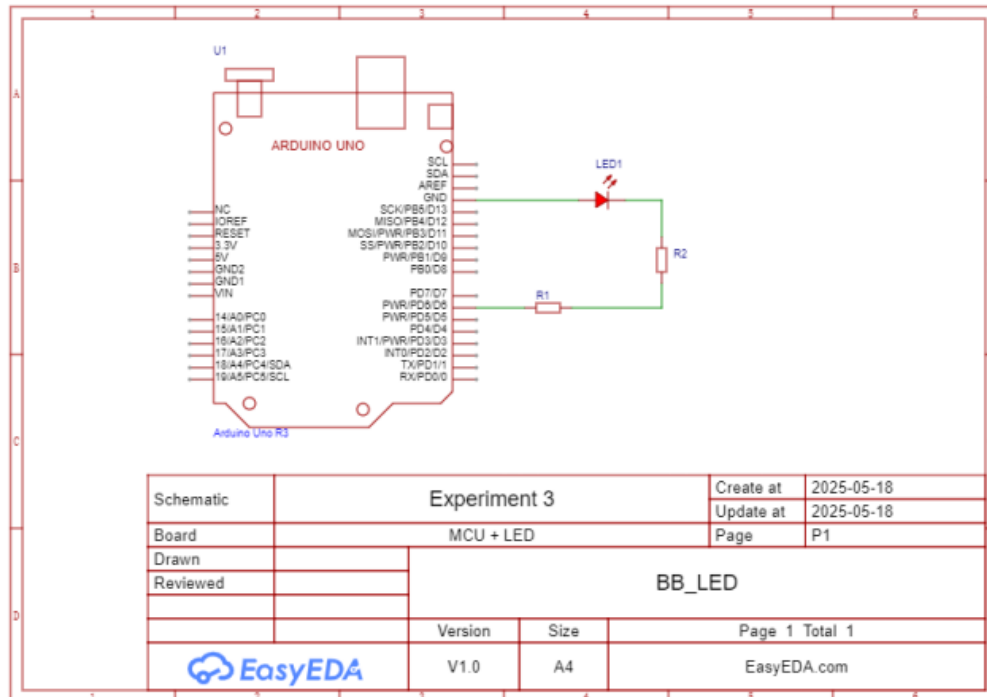
50% Brightness



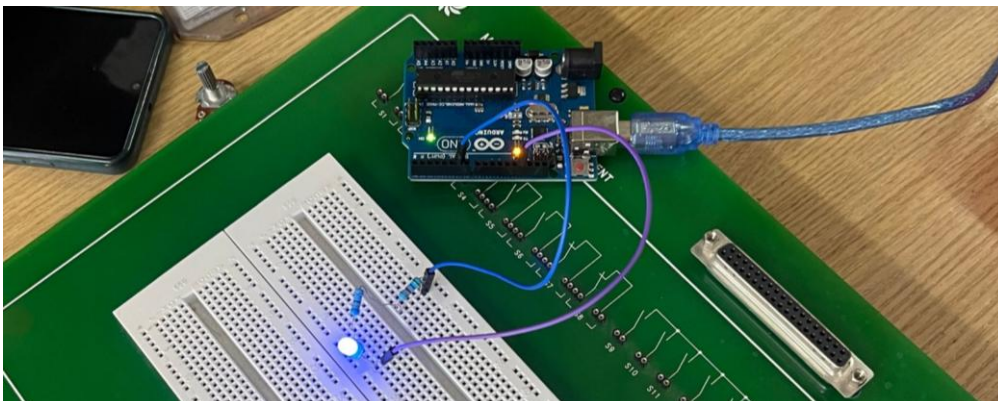
100% Brightness

Experiment 3

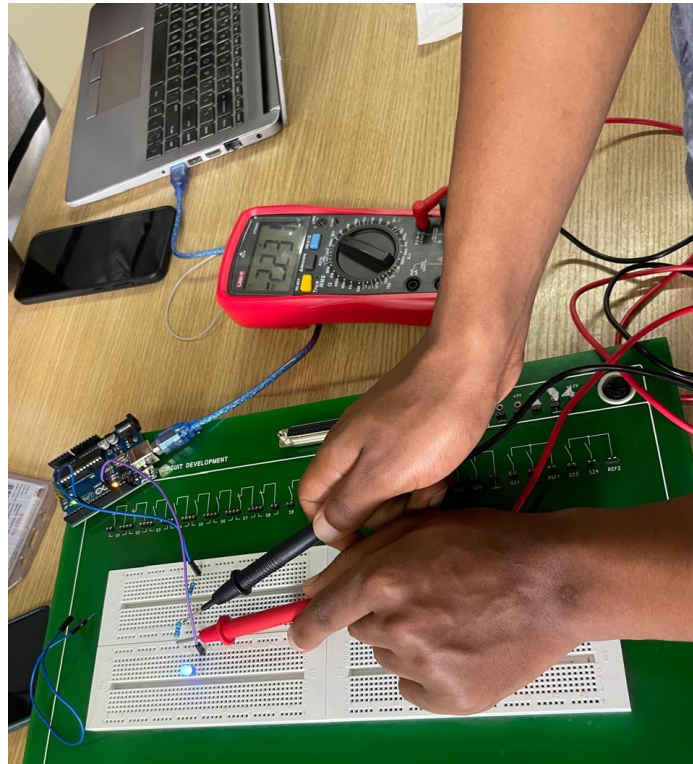
1. Using the breadboard, microcontroller, both resistors and LED, recreate the following schematic of resistors connected in series. Take a picture of the physical implementation.



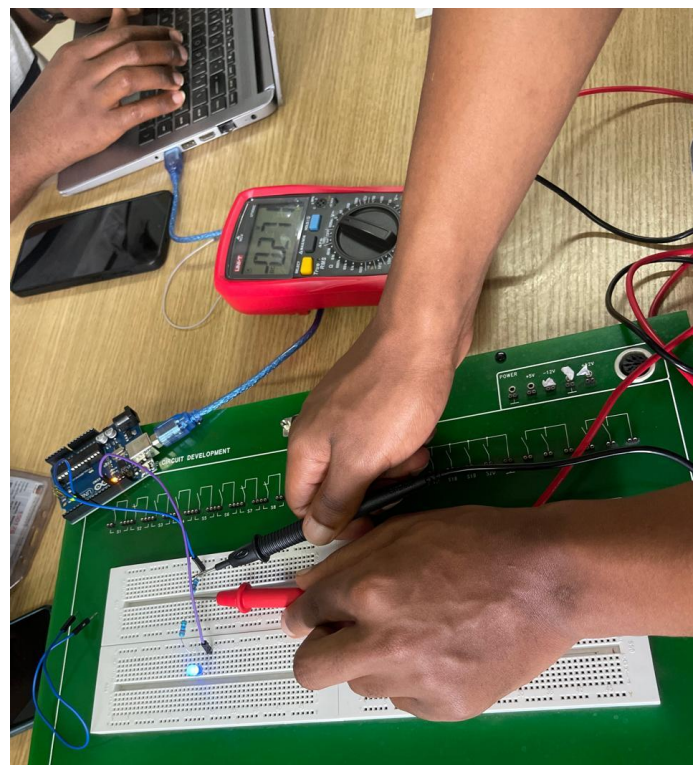
2. In Arduino IDE, re-deploy the [Blink](#) program and upload to your board.



3. Thereafter, use the multimeter to determine the voltage passing through the circuit. Write down the value registered at each resistor.



Measured Voltage, $V_1 = 2.231 \text{ V}$



Measured Voltage, $V_2 = 0.271 \text{ V}$

Experiment Reflection: Use Ohm's Law to estimate the current passed at each resistor.

$$V = IR$$

$$I_1 = V_1/R_1 = 2.231/2200 = \mathbf{0.0010141\ A}$$

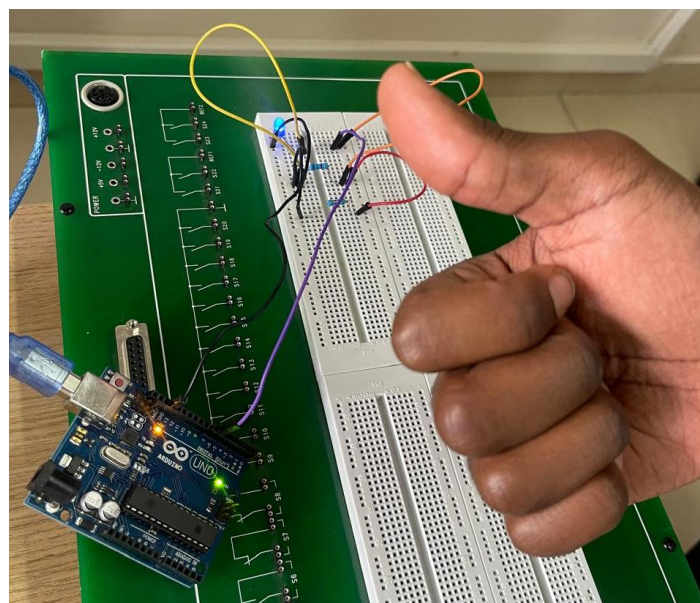
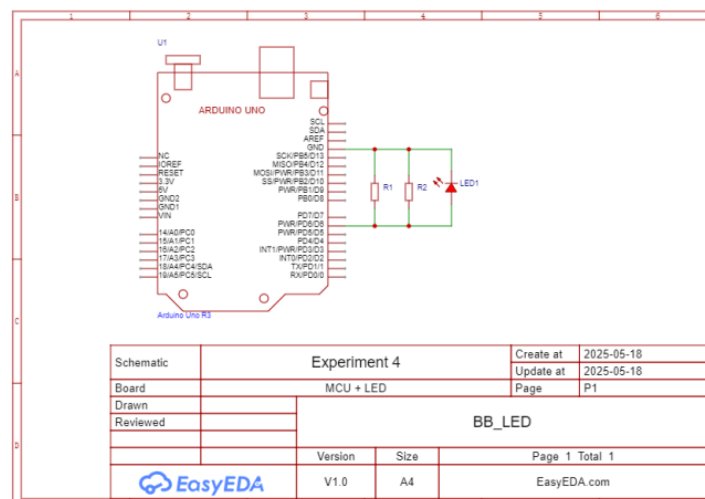
$$I_2 = V_2/R_2 = 0.271/270 = \mathbf{0.0010037\ A}$$

$$I \approx I_1 \approx I_2$$

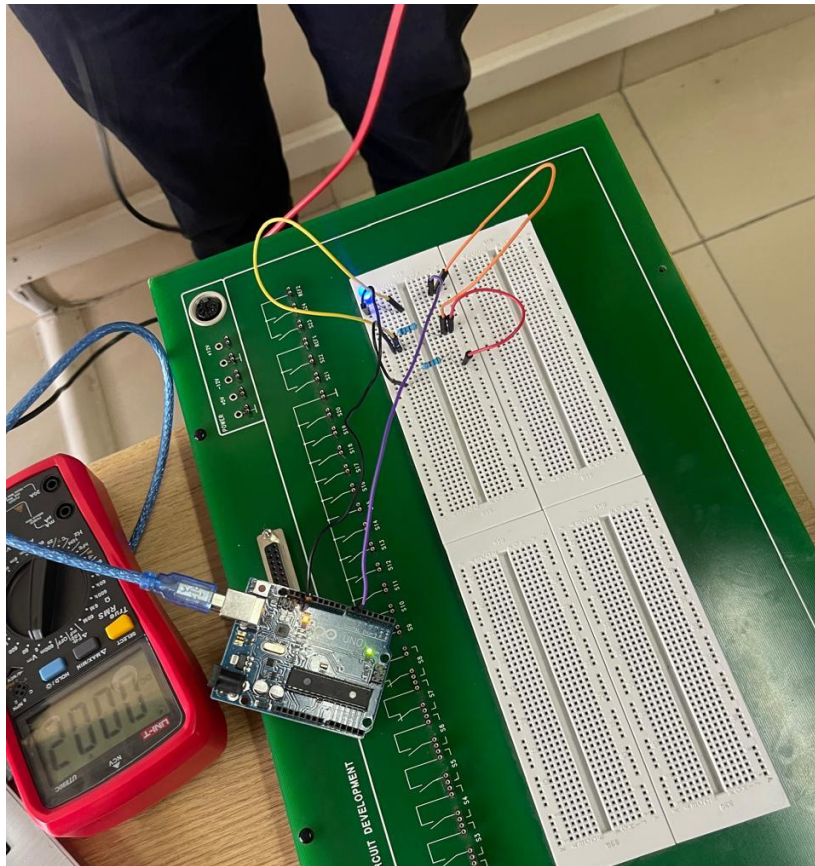
In a circuit where resistors are connected in series, the current remains constant. The above values are approximately equal, confirming that $I \approx I_1 \approx I_2$. Slight differences arise because of multimeter tolerance and potential human error when measuring values.

Experiment 4

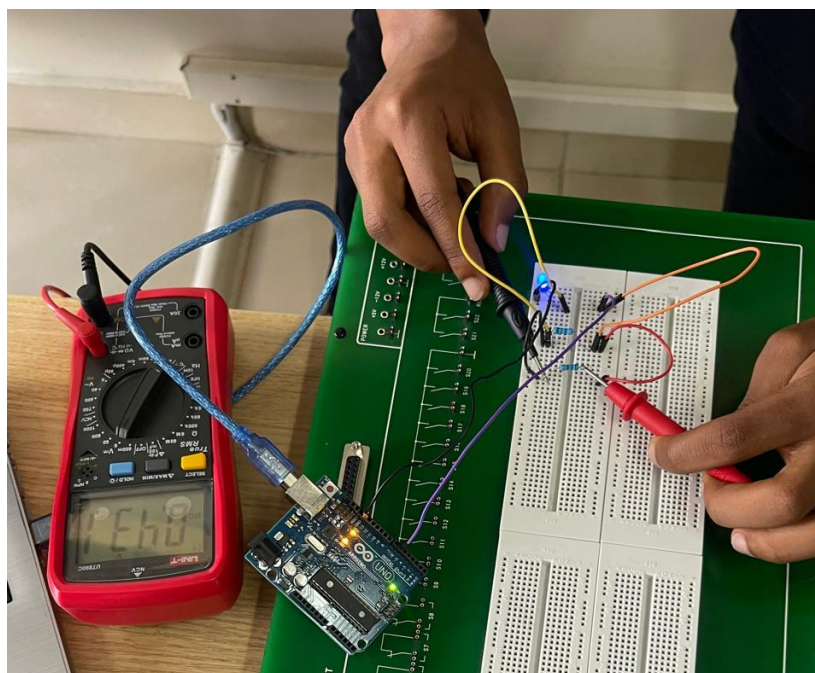
1. Using the breadboard, microcontroller, both resistors and LED, recreate the following schematic of resistors connected in parallel. Take a picture of the physical implementation.



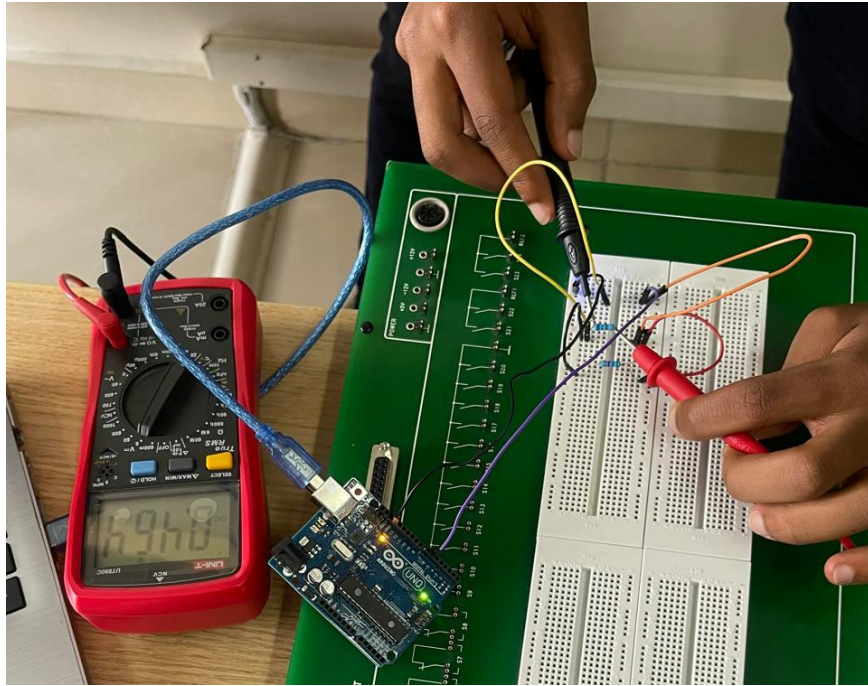
2. In Arduino IDE, re-deploy the [Blink](#) program and upload to your board.



3. Thereafter, use the multimeter to determine the voltage passing through the circuit. Write down the value registered at each resistor.



Measured Voltage, $V_1 = 0.431 \text{ V}$



Measure Voltage, $V_2 = \mathbf{0.464\text{ V}}$

For a circuit whereby resistors are connected in parallel, voltage is equal for each resistor.

Differences arise because of multimeter tolerance. Picking V_1 as our value for voltage;

Experiment Reflection: Use Ohm's Law to estimate the current passed at each resistor.

$$V = IR$$

$$I_1 = V_1/R_1 = 0.431/2200 = \mathbf{0.00019591\text{ A}}$$

$$I_2 = V_1/R_2 = 0.431/270 = \mathbf{0.0015963\text{ A}}$$

In a parallel circuit, voltage is equal across each resistor, with slight measurement differences due to multimeter tolerance and human error. Using Ohm's Law, $I_1 = 0.00019591\text{ A}$ and $I_2 = 0.0015963\text{ A}$. The higher current through R_2 ($270\ \Omega$) confirms that lower resistance allows more current to flow.

Meet the Team!

